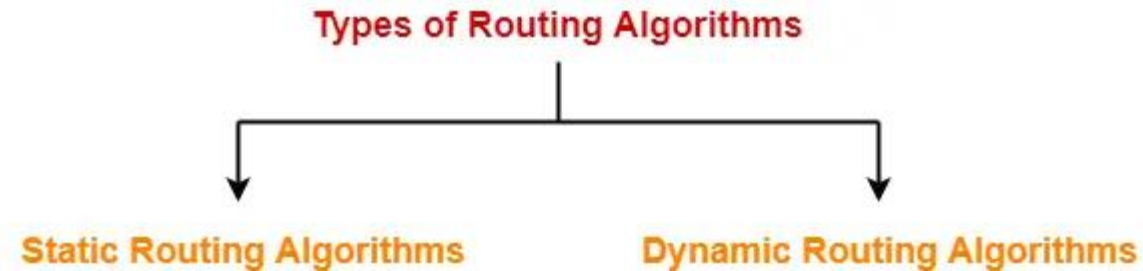


Routing Algorithms

Routing Algorithms

- Routing algorithms are meant for determining the routing of packets in a node.



Distance Vector Routing Algorithm

- Distance Vector Routing is a dynamic routing algorithm.

Step-01:

also known as Bellman Ford Algorithm or Ford Fulkerson

- Each router prepares its routing table. By their local knowledge. each router knows about-
- All the routers present in the network
- Distance to its neighboring routers

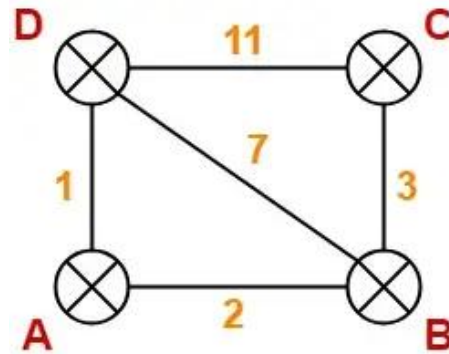
Step-02:

- Each router exchanges its distance vector with its neighboring routers.
- Each router prepares a new routing table using the distance vectors it has obtained from its neighbours.
- This step is repeated for $(n-2)$ times if there are n routers in the network.
- After this, routing tables converge / become stable.

Example

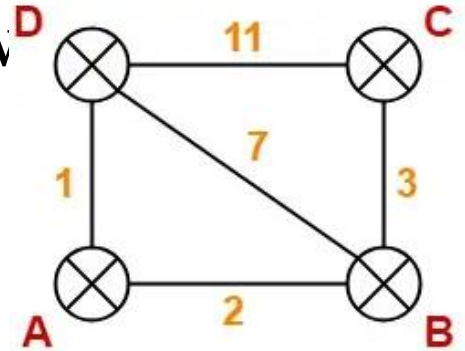
Consider

- There is a network consisting of 4 routers.
- The weights are mentioned on the edges.
- Weights could be distances or costs or delays.



Step-01:

- Each router prepares its routing table using its local knowledge.
- Routing table prepared by each router is shown below



At Router A

Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	–
D	1	D

At Router B

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	7	D

At Router C

Destination	Distance	Next Hop
A	∞	–
B	3	B
C	0	C
D	11	D

At Router D

Destination	Distance	Next Hop
A	1	A
B	7	B
C	11	C
D	0	D

Step-02:

- Each router exchanges its distance vector obtained in Step-01 with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.

At Router A-

$$D_x(y) = \min \{ C(x,v) + D_v(y) \} \text{ for each node } y \in N$$

- Router A receives distance vectors from its neighbors B and D.

From B

2
0
3
7

From D

1
7
11
0

Destination	Distance	Next hop
A	0	A
B		
C		
D		

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+7 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+11 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+7, 1+0 \} = 1$ via D.

Cost(A→B) = 2

Cost(A→D) = 1

New Routing Table at Router A

Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

Explanation For Destination B

- Router A can reach the destination router B via its neighbor B or neighbor D.
- It chooses the path which gives the minimum cost.
- Cost of reaching router B from router A via neighbor B = Cost (A→B) + Cost (B→B) = 2 + 0 = 2
- Cost of reaching router B from router A via neighbor D = Cost (A→D) + Cost (D→B) = 1 + 7 = 8
- Since the cost is minimum via neighbor B, so router A chooses the path via B.
- It creates an entry (2, B) for destination B in its new routing table.
- Similarly, we calculate the shortest path distance to each destination router at every router.

At Router B-

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-

From A	From C	From D	
0	∞	1	
2	3	7	
∞	0	11	
1	11	0	
Cost (B→A) = 2	Cost (B→C) = 3	Cost (B→D) = 7	New Routing Table at Router B

- Cost of reaching destination A from router B = $\min \{ 2+0, 3+\infty, 7+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+\infty, 3+0, 7+11 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1, 3+11, 7+0 \} = 3$ via A.

Thus, the new routing table at router B is-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-

From B

2
0
3
7

From D

1
7
11
0

Cost (C→B) = 3

Cost (C→D) = 11

Destination	Distance	Next hop
A		
B		
C	0	C
D		

New Routing Table at Router C

- Cost of reaching destination A from router C = $\min \{ 3+2, 11+1 \} = 5$ via B.
- Cost of reaching destination B from router C = $\min \{ 3+0, 11+7 \} = 3$ via B.
- Cost of reaching destination D from router C = $\min \{ 3+7, 11+0 \} = 10$ via B.

Thus, the new routing table at router C is-

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	10	B

At Router D-

- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-

From A	From B	From C	
0	2	∞	
2	0	3	
∞	3	0	
1	7	11	
Cost (D→A) = 1	Cost (D→B) = 7	Cost (D→C) = 11	New Routing Table at Router D

- Cost of reaching destination A from router D = $\min \{ 1+0, 7+2, 11+\infty \} = 1$ via A.
- Cost of reaching destination B from router D = $\min \{ 1+2, 7+0, 11+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+\infty, 7+3, 11+0 \} = 10$ via B.

Thus, the new routing table at router D is-

Destination	Distance	Next Hop
A	1	A
B	3	A
C	10	B
D	0	D

Step-03:

- Each router exchanges its distance vector obtained in Step-02 with its neighboring routers.
- After exchanging the distance vectors, each router prepares a new routing table.

At Router A

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B

2
0
3
3

From D

1
3
10
0

Cost(A→B) = 2

Cost(A→D) = 1

Destination	Distance	Next hop
A	0	A
B		
C		
D		

New Routing Table at Router A

Thus, the new routing table at router A is

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+3 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+10 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+3, 1+0 \} = 1$ via D.

At Router B-

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as

From A

0
2
5
1

Cost (B→A) = 2

From C

5
3
0
10

Cost (B→C) = 3

From D

1
3
10
0

Cost (B→D) = 3

Destination	Distance	Next hop
A		
B	0	B
C		
D		

New Routing Table at Router B

- Cost of reaching destination A from router B = $\min \{ 2+0 , 3+5 , 3+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+5 , 3+0 , 3+10 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1 , 3+10 , 3+0 \} = 3$ via A.

Thus, the new routing table at router B is

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as

From B

2
0
3
3

Cost (C→B) = 3

From D

1
3
10
0

Cost (C→D) = 10

Destination	Distance	Next hop
A		
B		
C	0	C
D		

New Routing Table at Router C

- Cost of reaching destination A from router C = $\min \{ 3+2 , 10+1 \} = 5$ via B.
- Cost of reaching destination B from router C = $\min \{ 3+0 , 10+3 \} = 3$ via B.
- Cost of reaching destination D from router C = $\min \{ 3+3 , 10+0 \} = 6$ via B.

Thus, the new routing table at router C is

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	6	B

At Router D

- Router D receives distance vectors from its neighbors A, B and C.

Router D prepares a new routing table as follows

From A

0
2
5
1

Cost (D→A) = 1

From B

2
0
3
3

Cost (D→B) = 3

From C

5
3
0
10

Cost (D→C) = 10

Destination	Distance	Next hop
A		
B		
C		
D	0	D

New Routing Table at Router D

- Cost of reaching destination A from router D = $\min \{ 1+0, 3+2, 10+5 \} = 1$ via A.
- Cost of reaching destination B from router D = $\min \{ 1+2, 3+0, 10+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+5, 3+3, 10+0 \} = 6$ via A.

Thus, the new routing table at router D is

Destination	Distance	Next Hop
A	1	A
B	3	A
C	6	A
D	0	D

Identifying Unused Links

After routing tables converge (becomes stable),

- Some of the links connecting the routers may never be used.
- In the above example, we can identify the unused links as-

We have

- The value of next hop in the final routing table of router A suggests that only edges AB and AD are used.
- The value of next hop in the final routing table of router B suggests that only edges BA and BC are used.
- The value of next hop in the final routing table of router C suggests that only edge CB is used.
- The value of next hop in the final routing table of router D suggests that only edge DA is used.

Thus, edges BD and CD are never used.

Important Notes

Note-01:

In Distance Vector Routing,

- Only distance vectors are exchanged.
- “Next hop” values are not exchanged.
- This is because it results in exchanging the large amount of data which consumes more bandwidth.

Note-02:

- While preparing a new routing table-
- A router takes into consideration only the distance vectors it has obtained from its neighboring routers.
- It does not take into consideration its old routing table.

Note-03:

- The algorithm is called so because-
- It involves exchanging of distance vectors between the routers.
- Distance vector is nothing but an array of distances

Note-04:

- The algorithm keeps on repeating periodically and never stops.
- This is to update the shortest path in case any link goes down or topology changes.

Note-05:

- Routing tables are prepared total $(n-1)$ times if there are n routers in the given network.
- This is because shortest path between any 2 nodes contains at most $n-1$ edges if there are n nodes in the graph.

Note-06:

- Distance Vector Routing suffers from count to infinity problem.
- Distance Vector Routing uses UDP at transport layer.